IIG_AWS_Analyzer Tool: User Manual

Version: 3.0
Target Hardware: Campbell Scientific CR1000/CR1000X
Output: Automated Plots for all Parameters (PNG & Interactive Plots)
Developed By- Varun Dongre, 45Th ISEA

1. System Requirements

Before running the tool on a new machine/PC, ensure the following are installed:

- **Python 3.10 or higher:** Download here (Ensure you check the box **"Add Python to PATH"** during installation).
- **Required Libraries:** Open your terminal (CMD) and run:
- pip install pandas matplotlib

2. File Preparation

The tool is designed to work with ASCII **TOA5** formatted files (.dat or .csv). But when you transfer files from CR1000 data logger it gives you binary files which needs to be converted into ASCII files using Card Converter tool. Once you convert all binary files into ASCII files this python script can be used for plotting.

Step 1: Launching the Program

Double-click the **IIG_AWS_Analyzer Tool.py** file or right click to open with python. A black command window will open and will ask to provide absolute folder path where all AWS data files are stored

D:\45ISEA\45ISEAData\AWS\March2026

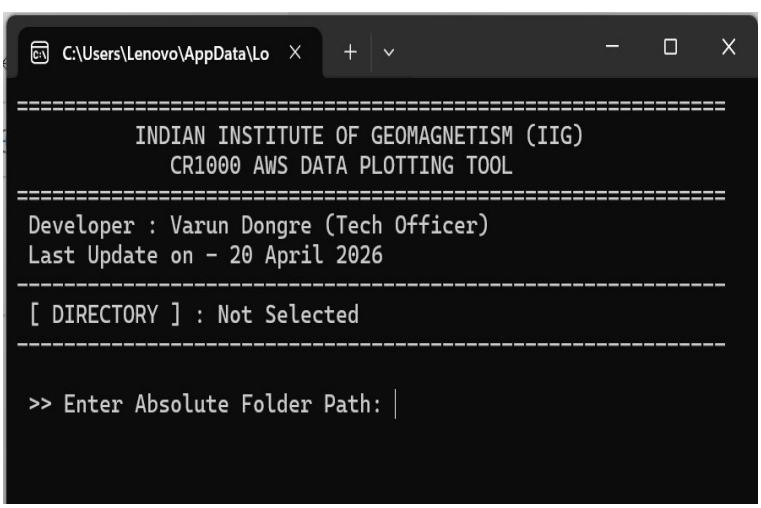

Figure 1 Tool Command Window

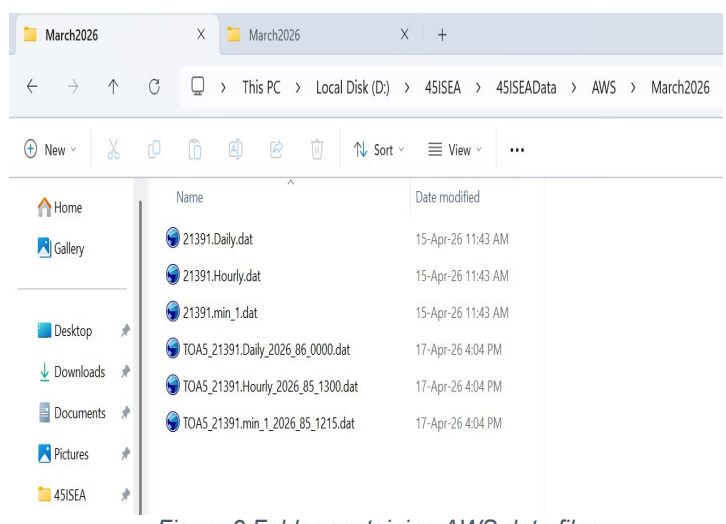

Figure 2 Folder containing AWS data files

Once you provide the correct folder path the tool will automatically detect all data Ascii (.dat & .csv) files available in that folder. Remember the tool will not parse raw binary files from CR1000 data logger (`eg: 21391.Daily.dat`)

```
C:\Users\Lenovo\AppData\Lo
==============================================================
          INDIAN INSTITUTE OF GEOMAGNETISM (IIG)
             CR1000 AWS DATA PLOTTING TOOL
==============================================================
 Developer : Varun Dongre (Tech Officer)
 Last Update on - 20 April 2026
--------------------------------------------------------------
 [ DIRECTORY ] : D:\45ISEA\45ISEAData\AWS\March2026
--------------------------------------------------------------

 [ INDEX ]    [ AVAILABLE AWS FILES ]
  0           21391.Daily.dat
  1           21391.Hourly.dat
  2           21391.min_1.dat
  3           TOA5_21391.Daily_2026_86_0000.dat
  4           TOA5_21391.Hourly_2026_85_1300.dat
  5           TOA5_21391.min_1_2026_85_1215.dat
  C           Change Directory
  Q           Quit

 >> Selection:
```

Step 2: Selecting the Data Source

- Type the **Index Number** (e.g., 0,1,2,3) of the file you wish to analyze & plot.
- Press **Enter**.
- **The tool will ask** you to select the Moving average filter, This filter is used to apply a smothering filter to smooth the noisy/jittery data. If you want to plot raw data just hit enter or enter the corresponding index number of the moving average point.

```
C:\Users\Lenovo\AppData\Lo
==============================================================
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 [ INDEX ]    [ AVAILABLE AWS FILES ]
  0           21391.Daily.dat
  1           21391.Hourly.dat
  2           21391.min_1.dat
  3           TOA5_21391.Daily_2026_86_0000.dat
  4           TOA5_21391.Hourly_2026_85_1300.dat
  5           TOA5_21391.min_1_2026_85_1215.dat
  C           Change Directory
  Q           Quit

 >> Selection: 3
    [MOVING AVERAGE FILTERING ]
[0] Raw [1] 3-pt [2] 5-pt [3] 10-pt
 >> Select Filter Strength: |
```

Step 3: Defining the Plot Period

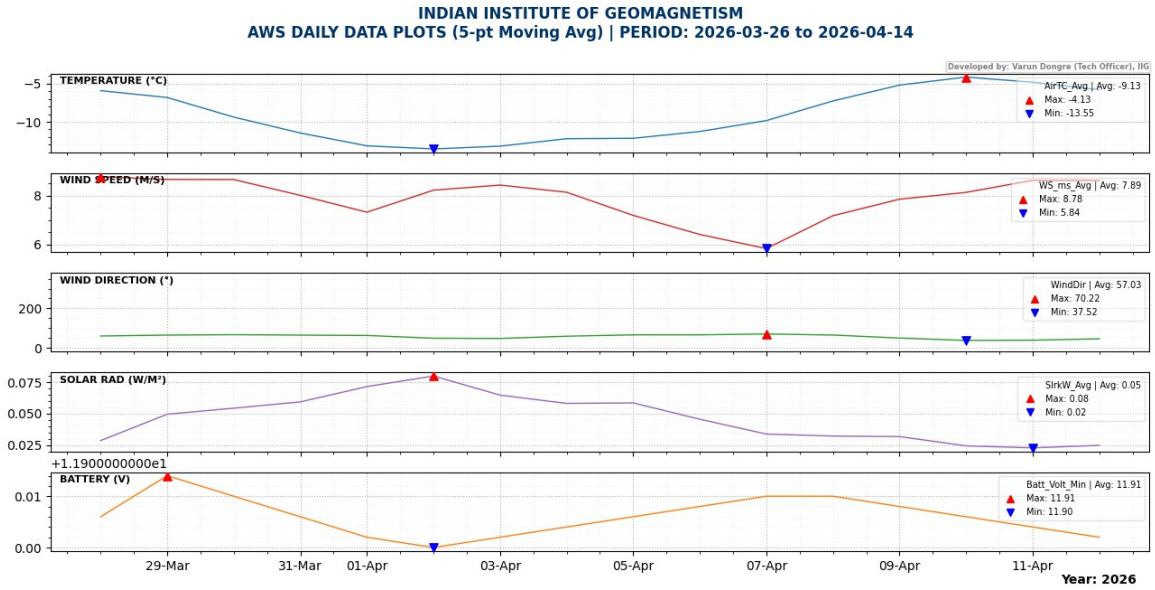
Once you select the moving average index number the tool will tell you the data range available in the file and you can choose data range to be plotted.

- **Format:** `YYYY-MM-DD` (e.g., `2023-10-20`)
- **Shortcut:** If you want to plot the **entire range**, simply press **Enter** for both prompts.

```
C:\Users\Lenovo\AppData\Lo

============================================================
          INDIAN INSTITUTE OF GEOMAGNETISM (IIG)
             CR1000 AWS DATA PLOTTING TOOL
============================================================
 Developer : Varun Dongre (Tech Officer)
 Last Update on - 20 April 2026
------------------------------------------------------------
 [ DIRECTORY ] : D:\45ISEA\45ISEAData\AWS\March2026
------------------------------------------------------------

 [ INDEX ]   [ AVAILABLE AWS FILES ]
  0          21391.Daily.dat
  1          21391.Hourly.dat
  2          21391.min_1.dat
  3          TOA5_21391.Daily_2026_86_0000.dat
  4          TOA5_21391.Hourly_2026_85_1300.dat
  5          TOA5_21391.min_1_2026_85_1215.dat
  C          Change Directory
  Q          Quit

 >> Selection: 3
    [MOVING AVERAGE FILTERING ]
[0] Raw [1] 3-pt [2] 5-pt [3] 10-pt
 >> Select Filter Strength: 2

     [ DATA AVAILABLE FROM ]
2026-03-26 00:00:00 to 2026-04-14 00:00:00
 >> Plot Start (YYYY-MM-DD): 2026-03-26
 >> Plot End   (YYYY-MM-DD): 2026-04-14
```

Step 4: Analyzing the Output

A high-resolution plot window will appear with separated subplots, minimum, maximum data points and proper file name. The plot window is an interactive window where you can zoom, select data point, export data to csv and finally save as png file.



Step 5: Changing folder path

- To analyze another file: Close the Plot window. The menu will reappear.
- To change the folder path type c and hit enter, it will re initiate to step 1.

Step 6 : Exiting the Tool

- To exit: Type `q` in the command window and hit enter or simply close the command window.

20-04-2026

End of the User Manual